

Gradient Descent: Core Notes

Gradient descent is an iterative optimisation algorithm that minimises a loss function by repeatedly stepping in the direction of the negative gradient.

The learning rate controls the size of each step. A rate that is too large can cause the optimisation to diverge, while a rate that is too small makes convergence painfully slow.

Batch gradient descent computes the gradient over the entire training set. Stochastic gradient descent uses a single example per step, which introduces noise but often converges faster in practice.

Momentum and Adaptive Methods

Momentum accumulates an exponentially decaying average of past gradients. This damps oscillation across ravines and accelerates movement along consistent directions.

Nesterov momentum looks ahead by evaluating the gradient at the approximate future position rather than the current one.

AdaGrad scales the learning rate per parameter by the square root of the sum of squared historical gradients, which suits sparse features but decays the rate aggressively over time.

Adam combines momentum with adaptive scaling and is frequently the default choice for deep networks.

Regularisation and Generalisation

Overfitting occurs when a model captures noise in the training set rather than the underlying signal, and so performs poorly on unseen data.

L2 regularisation penalises the squared magnitude of weights, encouraging smaller parameters and a smoother decision boundary.

Dropout randomly deactivates units during training, which prevents the network from becoming overly reliant on any single pathway.

The bias-variance tradeoff describes the tension between a model that is too simple to capture the structure and one that is too flexible to generalise.